Can Next-Token Oracles do Theory?

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Question. Suppose I gave you an oracle that can solve the Boolean satisfiability problem perfectly. What other cool things could you do with it?

To a practical person, such a question might seem silly: satisfiability oracles don't exist, and (probably) never will. So why should we entertain such nonsense?

Yet in the early 1970s, Cook, Levin, and Karp asked this question. The result was the theory of NP-completeness. They did this by using the notion of *reductions* between problems: methods of solving one problem, by using an oracle for the other problem. Importantly, they limited themselves to using the oracle as a black-box, where only its outputs were allowed, a restriction that yielded the generality of their results. Together their work showed that a wide variety of disparate tasks–integer programming, bin packing, job sequencing–were intimately connected. This had many far-reaching implications that echoed well beyond the statements.

We propose to ask the following:

Question. Suppose I gave you an oracle that can predict internet data optimally and flawlessly. What other cool things could you do with it?

As with the satisfiability oracle, we encourage the reader to refrain from intuitions born from existing machinery, nor the plausibility of its existence. The black-box nature of the thought experiment is essential to the generality of the ideas.

1 Next-Token Reductions

For those inclined, a first application would be to simply use it as a piggy bank. All manner of stock prices are on the internet, so one could simply point it at these, with a prompt like:

`Wall Street Journal, June 10 2026. The price of BTC hit` ...

But money is unimportant to a theorist. We would be more interested in having it finish our papers for us. Scientific papers abound on the internet, so our oracle's powers would apply to those. This is also true for the results sections of empirical papers. For instance it could help with:

`However, after we halved our learning rate` λ`, our test accuracy` ...

Because the vast majority of papers report the results they observed in the physical world, the true number is the most likely continuation, so the oracle must also predict the processes that generated that number. This applies equally well to any phenomena in the natural world:

`James Webb Telescope, Observation of Venus:` ...

`Artemis Mission Report:` ...

`Large Hadron Collider, 2012, Data Dump:` ...

An important note about these examples, is that humans would *not* be able to predict these things, yet the oracle can. In other words, despite its abilities being restricted to human-generated data, the oracle's capabilities are not bounded by those of humans. This is because, in reporting the LHC's data dump, humans never had to *generate* the sequence itself, but merely report what the physical universe was generating.

Yet CS theorists are more interested in *conceptual* research. But this is also within the oracle's purview. That means we can take any proof we're working on, give the oracle the first half, and let it continue the sequence:

> `Theorem.` If $f \in \mathsf{TF}_{l(n)}$, where $l(n) \geq (2n + 2\log n)$, is p-random, then f is one-to-one almost everywhere.
> `Proof.` Assume, for contradiction, that $f \in \mathsf{TF}_{l(n)}$ is p-random, $l(n) \geq (2n+2\log n)$, and there are infinitely many lengths n for which f has a collision. That is, for infinitely many lengths n, there exist distinct $x, x' \in \{0,1\}^n$ such that $f(x) = f(x')$. We will define

As we become accustomed to using it, with the instant query responses a true oracle provides, we realize much of our work is unnecessary. We could simply give it the theorem statement:

> `Theorem.` If $f \in \mathsf{TF}_{l(n)}$, where $l(n) \geq (2n + 2\log n)$, is p-random, then f is one-to-one almost everywhere.
> `Proof.`

And after enough rounds of this, we become more ambitious:

> `Theorem.` $P \neq NP$.
> `Proof.`

At this point, our oracle returns nonsense, and we suspect we've finally reached the limits of its powers.

But wait! Our oracle is not constrained by human expectations of what's possible–that's what makes it an oracle. But its superpower is to predict internet text, whether or not its content is "true". The most likely continuation of such a prompt is *not* a solution to a Millennium Prize Problem. A more frequent next token will be the beginning of a crank proof, or the setup for a joke; or most of all, a crankish, unserious paper on next-token prediction.

The problem is not with the oracle, but with the prompt we gave it. In the limit of an infinite internet, made up of data produced by humans over time, the frequency of actual separation proofs is likely to be nonzero. However, it is so vastly dwarfed by incorrect or irrelevant continuations. But this is a ultimately a constant difference, which can be made up for with more thoughtful prompts that condition on the former more than the latter. This may look something like:

```
Annals of Mathematics, July 2038
Lower Bounds for NTIME
Ryan Williams
Abstract.
```

We would further augment this with likely metadata from the journal, or even preface it with a brief news story, all without having to guess at the contents of the proof. Of course, we cannot estimate the relative frequencies between authentic and wrong continuations. The former may, after all, be 0.

But even if it is very small, there are many other ways of driving its conditional probability higher. For example, we could make n educated guesses about the high-level approach, or the lemmas in between, all checked instantly with the oracle as soon as we can write them. But notice that for large n, we're just doing standard complexity research, just with a sledgehammer that can instantly solve most sub-Millennium problems!

2 The Complexity Class of Next-Token Prediction

When I explain complexity classes like NP to my students, I often include examples of problems *not* in the class to delineate its limits. NP is vast, but $HALT$ and QBF (probably) show that its boundaries are not limitless. If it were ALL, then much of the discussion would be vacuous!

By the above examples, and others we could easily construct, it is hopefully clear that the theoretical power of next-token prediction is vast. We can define an informal complexity class by considering its closure (lower cone) under these oracle reductions. What does this class *not* contain?

Per the above discussion, it can predict any data on an infinite internet, the only restriction being that the internet was generated by humans over an infinite time period.

Then there are certain problems it cannot solve, such as:

```
Here's the hash of the message: 12tn4cy9n0c12n19c28.
And here's the message that was hashed:
```

We can of course name many other examples of problems it cannot solve, due to known or conjectured lower bounds in computability and complexity theory. A similar class of examples comes from:

```
I flipped a coin 10 times, here are the results:
```

Or more precisely, we could formulate the above in terms of true (quantum) randomness.

These examples also show that a *perfectly* accurate next-token predictor is impossible, since there are certain sequences that can't be predicted with perfect accuracy. An even simpler example is the empty prompt: what continues that? There exists some irreducible error even with infinite data.

For this reason, we've previously just considered *optimal* internet predictor, which does the best possible given the entropy of the data and other hard theoretical constraints. But as much as we could posit a Halting oracle, we can *consider* a 100% perfect predictor. What problems, if any, can it not solve?

Consider the problem of counting the number of galaxies outside the observable universe. This information can never physically get to humans, therefore no such data will ever appear on the internet with fidelity. The oracle will complete any such prompt, but the answers will have no correspondence to the true number.

This gives an interesting hierarchy of informal complexity classes:

1. Problems we can already solve with existing LLMs
2. Problems we could solve with an optimal next-token oracle
3. Problems we could solve with a perfect next-token oracle
4. Problems we couldn't solve *even with* a perfect next-token oracle

3 The Limits of Next-Token Prediction

We therefore see that next-token oracles can be very powerful. Their ultimate scientific potential is only limited to the following:

1. Any theoretical discovery humans could ever make, *if* we can name it

- For example, we cannot force the oracle to discover a novel philosophical idea if we cannot point to it in advance. On the other hand, we can name a hard conjecture such as P vs. NP as a goal. And although we can't name mathematical definitions in advance for it to discover, this is not such a limitation since the oracle would have to discover them in the process of proving hard conjectures.

2. Any empirical observation humans could ever report, *even if* they could never predict it themselves.

    - The above example of the Higgs boson showcases this, while the example of galaxies outside the observable universe shows the limitations.

Many computer scientists have assumed LLMs could never do science, because they are "just" next-token predictors. But by considering the black-box setting of language modeling, we can see that this is not true. *If* it is the case that LLMs can never resolve P vs NP, it may be due to the finiteness of pretraining data, or shortcomings of the SGD-based training process, or limitations of the transformer architecture. But the arguments above show the fault is not in the next-token task itself.

4 So can LLMs do Theory?

But these arguments *are* all theoretical: what do they actually show about the potential of real LLM-based systems?

We claim our notion of a "next-token oracle" does faithfully map to what LLMs can do, *in the limit of infinite pretraining data*. The "black-box" assumption reflects the following standard observations:

1. Large enough neural networks, including transformers, have the *capacity* to represent any function, however complicated.

    - This is essentially just the universal approximation theorem. In complexity theory terms, this is basically just the fact that large circuits can encode pretty much anything.

2. Stochastic gradient descent, if run for unlimited iterations on unlimited data on a fixed reward signal, is capable of optimizing neural networks to become arbitrarily capable at this task.

    - In the past, many people expected gradient descent wouldn't work because it would get trapped in local minima. However, we now know that this is vanishing unlikely in high-dimensional space, as one of many quirks where our low-dimensional intuitions were misleading. Moreover empirically, gradient descent has proved to be a robust workhorse that essentially never does this in practice.

This has an impressive upshot: standard LLMs, with infinite pretraining data, can do science at a superhuman level.

"But so what", you ask, "we don't actually have infinite data!" Moreover the really valuable prompt examples above are fairly sparse on the internet: there are only so many math papers or results sections of science papers. Hitting this data wall is probably why frontier labs have stopped focusing on pretraining.

But we haven't talked about RLVR! And if we squint at it from a distance, the automated production of math and coding tasks can be thought of as a means of arbitrarily augmenting the precious types of high-quality data that made pretraining worthwhile in the first place. It is a fraught process, subject to reward hacking and other difficulties the RL setting introduces.

In conclusion, we argue for a *task-centric view of AI capabilities*, where we first consider the narrow tasks they are trained on, and reason from there by means of black-box reductions. This task-centric view can be thought of as an extension of the compute-centric view, since in the regime of massive compute, the main question is whether this is put towards tasks that are useful (directly or via reductions to other tasks).